

Future of Market for Rooftop Solar PV Systems is Bright

With power crisis set to worsen in the near future, the potential for rooftop PV systems is huge in India. However, given the high upfront cost of the system, delay in subsidy disbursal and cheap imports, the path to growth may not be smooth for the solar system providers



UMA GUPTA

Electricity utilities are finding it difficult to meet the substantial growth in peak electricity demand in India. As a result, various industries and commercial establishments, and housing complexes in cities and towns, have to use diesel generators for power backup even during the day time. Rooftop solar PV systems offer them an opportunity to reduce energy costs and ensure power continuity in the event of grid unavailability. That is the reason why the use of solar/diesel hybrid rooftop applications is growing rapidly throughout India.

However, one cannot accurately quantify the size of the rooftop PV market. The market is just evolving.

"While many PV solutions for off-grid, rooftop, standalone applications have matured during the last 25 years in the country, the market is awakening to the reality of PV possessing every type of power generation solution that is modular and flexible for customisation to suit the needs of an individual. In the next few years, PV systems will receive greater acceptance by every segment of the market due to the heightened awareness," explains Sunil Goel, vice president, Maharishi Solar Technology.

With government subsidies, announced mainly by the southern states of Tamil Nadu, Andhra and Kerala, this market will grow faster over the next few years so long as subsidy can be realised by the end user.

In his presentation at Intersolar India 2012, Madhavan Nampoothiri, founder and director of RESolve Energy Consultants, shared the key findings of a KPMG report which said that rooftop power generation potential in India is expected to reach 4 GW by 2016-17. Grid parity could be achieved by 2015-16. Favourable trends for rooftop PV include increase in power tariffs in many states, transmission and distribution losses of about 20 per cent (which can be avoided by rooftop PV) and target achievable subject to creation of an enabling environment by regulators and utilities.

Rooftop vs multi-megawatt plants and home inverters

According to P.S. Deodhar, founder, MITRAMAX Energy, rooftop solar power generation is the best development. Multi-megawatt plants are a rather foolish option where utilities that buy the power waste over 30 per cent of that expensive solar energy in various losses that are integral to the Indian grid. In addition to that, there are losses in transformers that step up the solar inverter's 400V output to 66 kV. In the case of rooftop solar power option, however, users get good quality power during blackouts.